

TOPOLOGY-MARGINALIZED HARMONIC SOURCES FOR BOND-FREE MOLECULAR FLOW MATCHING

Anonymous authors

Internal development version

Internal research draft. Results are scoped to the reported benchmark.

ABSTRACT

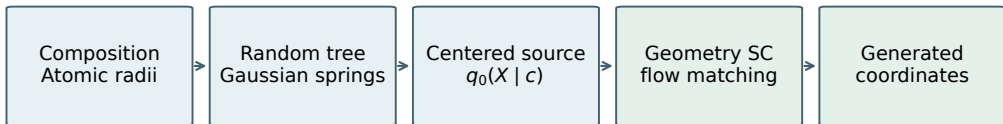
Generating molecular coordinates without a supplied bond graph requires a model to create both spatial organization and chemical compatibility. Isotropic noise ignores atomic scales, whereas graph-conditioned harmonic priors require the connectivity that this task leaves unspecified. We study a harmonic source that marginalizes over random dependence trees on the given atoms. Each component is a Gaussian spring network on the translation-free subspace, and a matrix-tree ratio gives its normalized marginal density. Atomic radii and simple species propensities set the source; the selected method adds no learned source head. We combine it with geometry self-conditioning in a bond-free molecular flow-matching model. In matched OMol25 continuations, harmonic sources improve chemical-graph support over isotropic Gaussian sources by 5.86 percentage points on 12 development compositions. Frozen models improve by 8.83 points on 10 additional reference-qualified compositions absent from two verified processed corpora, with a conditional paired 95% interval of [5.08,12.50]. Both continuation seeds improve at the same denoiser-call budget. A narrower shell source and several learned additions fail their controls. A calibrated covariance-Gaussian control also loses in both panels, supporting a role for mixture structure beyond the tested second-moment approximation. An energy-model readout also finds lower mean energy and force magnitude. These results support a useful molecular source design in a limited domain; competitiveness against independent generators and thermal-distribution accuracy require separate evidence. The results support incorporating latent spatial structure into a molecular flow without supplied bond labels or an additional learned source model. A separate exploratory physics-feedback test improves energy and force after paired parameter transfer, without establishing an extra benefit from work weighting.

1 INTRODUCTION

Molecular generation is difficult partly because local geometric constraints must emerge from an initially unstructured configuration. A starting distribution can reduce this burden before a neural transport is learned. Flow matching (Lipman et al., 2023) permits a choice of source distribution, and molecular conformer generators already exploit harmonic sources constructed from known chemical graphs (Hassan et al., 2024). Our setting instead provides atomic composition and electronic conditions but no chemical bonds. A prior requiring the target graph would change the task.

We investigate a source that integrates over possible spatial dependence trees. A sampled tree connects Gaussian edge displacements whose scales depend on atomic radii. Integrating these displacements and removing translation gives a configuration with correlated atomic positions. The latent edges describe the source construction; the generator receives coordinates and conditions, without the sampled tree, and the edges do not constrain its final chemical graph. A matrix-tree expression makes the source density explicit after the tree is marginalized. This combines established probabilistic ingredients in a molecular flow-matching source rather than changing the flow-matching loss.

The empirical question is whether this structure helps a capable generator. Geometry self-conditioning is therefore shared by all primary source comparisons. We first test isotropic Gaussian, harmonic-tree



Source trees are latent dependence structures; no supplied chemical bonds

Figure 1: Implemented generation path. Atomic radii and species propensities define a random harmonic network used to draw coordinates. The field receives the coordinates and fixed conditions, not the sampled tree. No energy reweighting is used in the reported generation results.

and narrow shell-tree sources with matched network weights at initialization, data rows, optimization and inference budgets. Harmonic trees perform best in both continuations. We then freeze the harmonic and Gaussian models and evaluate all remaining 10 qualified compositions from the reference pool. Their positive result motivates this candidate. A subsequent covariance-Gaussian control is also worse, supporting an effect beyond the tested covariance approximation. These results do not turn source normalization or self-conditioning into a novel theorem.

Our contribution is a task-specific probabilistic source construction, its integration with a molecular flow, and controlled evidence for its utility. We make the physics connection explicit through a Gaussian spring energy and separate this auxiliary source energy from an electronic potential. Flow time is an interpolation parameter, not laboratory time; learned paths are not asserted to follow molecular dynamics. A Jarzynski-type reweighting connection is retained with its assumptions in Appendix F; it is not used to infer equilibrium accuracy from geometric validity.

2 RELATED WORK AND SCOPE

Sources and molecular flows. FlowMol3 combines joint topology/coordinate generation with self-conditioning, fake atoms and geometry distortion (Dunn & Koes, 2025). We reuse its equivariant vector-field implementation; self-conditioning is established prior art. ET-Flow constructs harmonic sources for graph-conditioned conformers (Hassan et al., 2024). Its molecular input graph and benchmark differ from our composition-conditioned coordinate task. Source-Guided Flow Matching studies guidance by changing sources under a frozen field (Wang et al., 2026); condition-dependent source learning is also established (Kim et al., 2026). Thus neither changing noise nor learning a source in general is our novelty claim.

Latent trees. Factored distributions over trees and tractable tree learning are classical (Meila & Jaakkola, 2000; Duan & Dunson, 2023). Random-spanning-tree difference priors, including Gaussian mixtures and conjugate tree posteriors, have also been studied for imaging inverse problems (Everink, 2026). The molecular instantiation here uses three-dimensional isotropic edge displacements, an intrinsic centered coordinate measure, species-dependent scales, and no supplied molecular graph. These distinctions define the proposed application; strong molecular controls are still needed to establish its practical and scientific significance.

Target distribution and physics. OMol25 provides calculated structures and properties across electronic conditions (Levine et al., 2025); its empirical geometry distribution is not a specified-temperature canonical ensemble. Harmonic source energies encode spatial fluctuations, whereas an electronic potential describes different physics. Jarzynski’s relation and its single-molecule pulling applications connect nonequilibrium work to equilibrium free energies (Jarzynski, 1997; Hummer & Szabo, 2001). An arbitrary learned transport or higher graph-validity rate does not by itself supply the required work weights. We therefore separate our measured generation task from the optional energy-correction construction in Appendix F.

3 A HARMONIC SOURCE WITH LATENT CONNECTIVITY

Let $c = (z_1, \dots, z_N, Q, S)$ contain atomic numbers, total charge and spin multiplicity. Coordinates lie in $\mathcal{H}_N = \{X \in \mathbb{R}^{N \times 3} : \sum_i x_i = 0\}$, with intrinsic Lebesgue measure and dimension $3(N-1)$. For positive symmetric affinities a_{ij} , define

$$p(T | c) = \frac{\prod_{(i,j) \in T} a_{ij}}{\tau(a)}, \quad \tau(a) = \sum_{T \text{ spanning tree}} \prod_{(i,j) \in T} a_{ij}. \quad (1)$$

The selected fixed source uses $a_{ij} = u_i u_j$. We set $u_i = 0.05$ for H and F/Cl/Br/I, and otherwise $u_i = \max(1, \text{default valence}(z_i) - 1)$. This species heuristic discourages terminal atoms from becoming high-degree source hubs; it neither imposes nor predicts chemical bond orders. The fixed source uses species but not Q, S beyond validating their admissibility.

Gaussian edge laws and sampling. Let ℓ_{ij} be the sum of covalent radii and $\sigma_{ij}^2 = \ell_{ij}^2 \exp(s^2)/3$, with $s = 0.2$ fixed throughout. The edge-displacement law is

$$h_{ij}(d) = (2\pi\sigma_{ij}^2)^{-3/2} \exp\left[-\frac{\|d\|^2}{2\sigma_{ij}^2}\right]. \quad (2)$$

The scale convention matches the covariance of the previously tested lognormal shell edges; it was not optimized on the confirmation outputs. Sample a tree using weighted Wilson sampling, draw its independent edge vectors, integrate along the tree, and subtract the coordinate mean. The resulting nominal source is invariant under rotations and permutations of identical species.

Proposition 1 (Intrinsic mixture density). *For normalized isotropic edge densities,*

$$q_0(X | c) = N^{3/2} \frac{\tau(a_{ij} h_{ij}(x_i - x_j))}{\tau(a)} \quad (3)$$

is normalized on $\mathcal{H}_N$. For a log affinity $\alpha_e = \log a_e$,

$$\partial_{\alpha_e} \log q_0(X | c) = \Pr(e \in T | X, c) - \Pr(e \in T | c). \quad (4)$$

Appendix A applies standard incidence-matrix and tree-ensemble identities. Two grounded Laplacian determinants evaluate the density in cubic time; the implementation uses positive log-space elimination for numerical stability. This is a density marginalized over trees, not conditional on the particular tree used to sample X_0 .

What physics is applied? For a fixed tree, define the dimensionless spring energy

$$\mathcal{U}_T(X) = \frac{1}{2} \sum_{(i,j) \in T} \frac{\|x_i - x_j\|^2}{\sigma_{ij}^2}. \quad (5)$$

The component is a normalized Gaussian proportional to $e^{-\mathcal{U}_T}$. The marginal $-\log q_0$ is a soft mixture of these spring energies, with the component normalization constants retained. Its coordinate score is

$$\nabla_{x_i} \log q_0(X | c) = - \sum_{j \neq i} \Pr((i, j) \in T | X, c) \frac{x_i - x_j}{\sigma_{ij}^2}. \quad (6)$$

This gives a posterior-averaged elastic coupling. It follows by differentiating Equation (3), rather than constituting a new fluctuation law. The springs have zero rest displacement; covalent radii set their fluctuation scale, not an excluded-volume guarantee or physical bond-equilibrium potential. The evaluated sampler draws this source; it does not use Equation (6) as an additional learned force or guidance term.

4 INTEGRATION WITH MOLECULAR FLOW MATCHING

Sample $X_0 \sim q_0(\cdot | c)$ and a reference X_1 from the training corpus. A type-preserving assignment and proper rotation couple the pair, followed by a shared independent Haar augmentation.

This preserves the invariant source marginal while allowing dependence between paired endpoints (Appendix B). We use the linear path $X_t = (1 - t)X_0 + tX_1$ and target $V^* = X_1 - X_0$.

The FlowMol coordinate updater returns $D_\theta(X_t, t)$ and the research parameterization interprets $v_\theta = D_\theta - X_t$ as velocity. This explicitly trained displacement interpretation differs from a native endpoint head. A first pass yields $v^{(0)}$ and the endpoint estimate $\hat{X}_1 = X_t + (1 - t)v^{(0)}$. A second pass recycles that estimate through the existing geometry self-conditioning modules to obtain $v^{(1)}$. Both passes remain differentiable, with objective

$$\mathcal{L}_{\text{FM}} = \frac{1}{2} \mathbb{E} \left[\|v^{(0)} - V^*\|^2 + \|v^{(1)} - V^*\|^2 \right]. \quad (7)$$

Atomic categories and the declared charge interface remain fixed; edge inputs carry the constant no-bond interface. No bond-classification loss is introduced. The sampled tree is not supplied to the vector field. Self-conditioning and the endpoint-estimation identity are existing techniques shared by all primary arms.

Generation integrates $v^{(1)}$ with 32 midpoint steps and two passes per field call, totaling 128 primitive denoiser evaluations per attempted sample, followed by 0.025 Å centered Gaussian noise. We retain this numerical sampler exactly for comparisons. Source normalization alone does not give the finite integrator plus noise a qualified final density. The main experiment uses FM only; existing force/energy training interfaces are not evidence for a thermal law, and the independent-Gaussian velocity-to-score proxy is invalid for this source/coupling.

5 EXPERIMENTAL DESIGN

Task and training. We evaluate coordinate generation for neutral-singlet organic compositions, containing C and H and elements from H/B/C/N/O/F/Si/P/S/Cl/Br/I. The implementation supports 200 atoms; experiments are limited to 8–40. Two continuations share one pretrained warm checkpoint. Each uses 3,000 distinct OMol25 training rows and 3,000 optimizer updates, with identical rows and initial weights across source arms within that continuation. The source is fixed and adds no fitted parameters. The vector-field parameter count is 5,904,369. Backbone learning rate is $2 \cdot 10^{-5}$; activated self-conditioning modules use $3 \cdot 10^{-4}$. Two differentiable passes are used for every primary source arm.

The training subset is deliberately narrower than the full processed corpus: neutral-singlet organic references must pass the unchanged graph assay and admit the declared bounded-edge, degree-capped geometric scaffold used in an earlier training selection. The source-SC models reuse these coordinates but ignore the stored scaffold. All arms share this selection. It is derived geometric selection, not supplied bond supervision, and limits claims about unrestricted OMol25. Additional training details and excluded alternatives are retained in the appendix.

Panels and separation. Original reference qualification identified 31 accepted compositions among 78 metadata-eligible candidates. An earlier 21-composition panel used up to 8 per size bin; the present 12-composition development panel is a fixed subset of that panel. After comparing all three source arms, we selected harmonic sources jointly across the two continuations and froze both harmonic and Gaussian models. We then evaluated *all* remaining 10 qualified compositions. They happen to have 17–28 atoms; no generator output was used to filter them. Exact count-vector checks find no matching composition in 3,902,107 processed training rows or 39,415 validation rows. References qualify the assay and do not initialize the generator or enter fitting. Separation from these corpora does not certify unrecorded pretraining or all trajectory relations. The additional panel reuses the same trained models, not two further independent training replicates.

Metrics and uncertainty. Every method attempts 64 samples per composition. Geometry support requires contact connectivity and no short-distance overlap using 1.25 and 0.6 times summed covalent radii. Graph support additionally requires successful chemical-graph perception. This assay does not certify quantum stability, the complete electronic state or synthetic accessibility. Failed outputs and validator exceptions stay in the denominator. We also record distinct perceived connectivity within each composition and actual generation time. Paired bootstrap intervals condition on the fitted models and fixed compositions; a composition bootstrap separately summarizes heterogeneity. Neither is a broad population-generalization certificate.

Table 1: Chemical-graph support per attempted output. All primary arms share the geometry-SC backbone and 128 primitive denoiser calls. The additional panel uses frozen models selected after the development comparison.

Panel	Continuation	Gaussian	Shell tree	Harmonic tree
Development (12)	1	357/768	417/768	436/768
	2	377/768	342/768	388/768
Additional (10)	1	326/640	—	413/640
	2	349/640	—	375/640

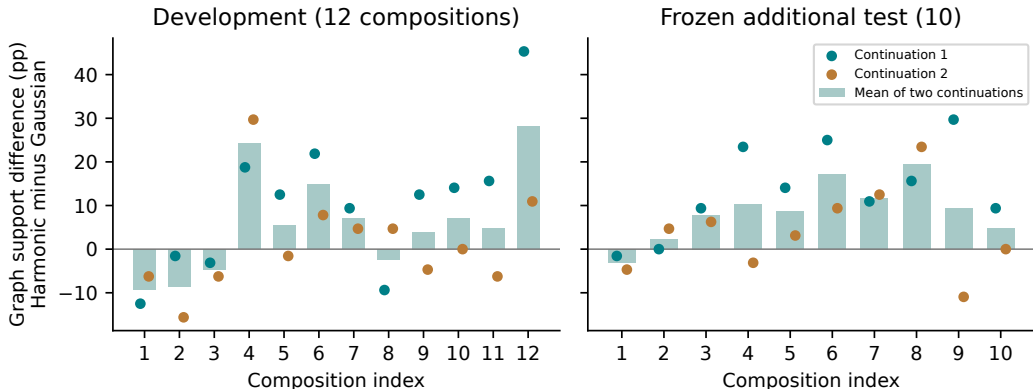


Figure 2: Harmonic-source graph-support gains vary across compositions and continuations. Points show individual continuation differences; bars average them. Positive aggregate gains do not imply improvement for every composition. The additional panel uses the same two models as development.

6 RESULTS

Harmonic trees improve graph support over isotropic Gaussian sources by 5.86 percentage points on the development panel, with conditional paired 95% interval [2.67,8.98] and descriptive composition interval [0.52,12.17]. Both continuation signs are positive, although the first gain is substantially larger. The narrow shell source fails its prespecified superiority gate and is worse than the harmonic tree by 4.23 points, with paired interval [-7.29,-1.11].

On the additional 10 compositions, harmonic sources improve graph support from 52.73% to 61.56%, or 8.83 points, with paired interval [5.08,12.50] and descriptive composition interval [4.84,12.66]. Geometry support improves by 9.69 points, with paired interval [6.33,12.89]. All graph-supported samples have distinct perceived connectivity within their composition in this panel. This is an assay-specific diversity readout; structural stereochemical novelty is not inferred from it. No validator exception occurs. Harmonic/Gaussian generation times are 175.73/175.43 seconds and 178.20/177.74 seconds for 640 samples. These measurements do not establish a speed advantage. All 2,560 additional-panel source and structural records replay.

Control for source covariance. A single Gaussian source uses the existing 128-tree estimate of the harmonic mixture covariance, symmetrized over identical species. Calibration against an independent 8,192-tree estimate gives 1.85–6.25% relative Frobenius error across all 22 compositions; this control is explicitly approximate. Its source settings were inherited from the earlier control implementation. It was added after the initial additional-panel result, with the same data, optimizer, network and sampling budgets; neither harmonic model was changed.

Harmonic sources improve graph support by 11.65 points on development (paired 95% [8.59,14.78]; descriptive composition [8.07,15.43]) and 15.39 points on the additional panel (paired 95% [11.64,19.06]; composition [11.95,18.98]). All four point differences are positive. The covariance Gaussian is also worse than the isotropic Gaussian in both panels. Thus simple covariance

Table 2: Covariance-Gaussian control at matched continuation and inference budgets. Its covariance is a finite-tree approximation, not an exact identity.

Panel	Continuation	Covariance Gaussian	Harmonic tree
Development	1	332/768	436/768
	2	313/768	388/768
Additional	1	286/640	413/640
	2	305/640	375/640

Table 3: Harmonic minus isotropic Gaussian physical readouts on the frozen additional panel. Lower is better for these surrogate-quality measures. Intervals are conditional paired 95% intervals.

Readout	Support	Difference	Interval
Energy (eV/atom)	All attempts	−0.0406	[−0.0579, −0.0232]
	Common graph	−0.0340	[−0.0692, −0.0003]
Force RMS (eV/Å)	All attempts	−0.4635	[−0.6782, −0.2512]
	Common graph	−0.5389	[−0.8950, −0.2089]

replacement does not explain the observed harmonic-source advantage in these continuations. Finite covariance-estimation error and model optimization remain limitations of an exact higher-moment attribution. All 2,816 new control outputs replay, and parameter counts, initial feedback weights, data rows and training recipes match the frozen references. This is a new control on examined panels, not another untouched test.

Energy-model readout of the frozen generators. We score every output on the ten-composition panel and each original reference with the same conservative eSEN checkpoint at its original charge and spin. To retain the parity convention of the earlier oracle audit, use $E_+(X) = [E(X) + E(-X)]/2$ and $F_+(X) = [F(X) - F(-X)]/2$. The model measures are an energy proxy and force consistency, not independent quantum validation. Differences are paired within composition before averaging; energies per atom also subtract that composition’s reference energy, which cancels in the paired difference. There are 5,160 raw energy/force queries, including both inversions of the references in each continuation’s readout. All stored inputs, outputs and derived quantities replay in the audit.

Force RMS means the square root of the per-atom mean squared force norm, then averaged over samples. The common-graph statistic uses only paired outputs accepted by both methods; cell counts range from 3 to 35. It is a selected-support estimand and is not an unbiased equilibrium expectation. All-attempt statistics include outputs rejected by the structural assay and may reflect potential-model extrapolation. The two readouts therefore answer different questions and are both retained. Their positive direction strengthens the quality interpretation of the source result, but lower energy or force magnitude does not establish a Boltzmann distribution or remove the need for independent physical validation.

Continuation effects are heterogeneous. All-attempt energy differences are −0.08049 and −0.00070 eV/atom; force-RMS differences are −1.21996 and +0.29302 eV/Å. Thus the pooled force advantage does not replicate in sign across both continuations, and the second energy effect is small. The conditional intervals do not quantify uncertainty over independently retrained models.

Original endpoint-checkpoint reference. We also evaluate the original 30,000-step OMol25 checkpoint using its endpoint semantics, 127 Euler updates and first-step bootstrap/history conditioning, for 128 actual denoiser calls. Current composition and the neutral/no-bond interface are clamped to define the same coordinate-generation task. Native predicted history gives 106/640 and 113/640 graph passes; a variant with fixed history categories and a centered first bootstrap gives 116/640 and 120/640. The two streams use the same original checkpoint, not two training replicates. All 2,560 outputs replay. The latter changes bootstrap centering as well as categorical history and is not a pure categorical-feedback ablation. This is a task-adapted reference, not the published joint-generation benchmark. The source-SC models have extra shared adaptation and 3,000 specialized

updates; these differences cannot be attributed solely to source choice. The matched Gaussian and covariance-Gaussian continuations remain the primary source controls.

What did not help. We retain failed source-coordinate likelihood learning, source-utility adaptation, static and dynamic tree context, fixed-tree product coordinates, and localized edge-feedback experiments in the appendix. Their failures prevent an extra learned-module claim. Geometry self-conditioning improves the original one-pass shell baseline by 5.47 points but is prior art. The harmonic-source result is a separate source comparison with that stronger mechanism shared by all arms.

Physics feedback in learning. We additionally test a local force-escorted teacher with complete Jarzynski work weights and matched unweighted/replay controls (Appendix G). Complete work improves physical readouts relative to its matched unweighted student, but all direct continuations harm the frozen generator. A subsequent fixed-coefficient parameter-difference test largely restores observed graph support while reducing energy and force in both continuations. Complete-work and unweighted versions perform similarly after transfer; a Jarzynski-specific extra benefit is not established. This is exploratory evidence on an examined panel, not an independent confirmation or a global Boltzmann claim.

7 DISCUSSION AND LIMITATIONS

A simple fixed probabilistic source now has a positive controlled result and a positive additional-composition test. The useful candidate is a random harmonic network, not the initially proposed narrow shell or a new learned affinity head. This shifts the method story toward spatial source design with latent connectivity. Normalization, posterior elastic coupling and symmetry give a clear mathematical specification, but the underlying tree identities and Gaussian spring physics are established. The contribution must be judged by its specific molecular instantiation and reproducible utility.

Coverage remains small, models share a warm start, and the target is a selected neutral-organic subset. A benefit at this continuation budget does not establish an asymptotic advantage after fully converged training. Native FlowMol3 unconditional validity and ET-Flow conditional-conformer metrics cannot be substituted for a matched generator comparison. The completed original-endpoint comparison is task-adapted and has a different training history. We do not claim superiority over those full published systems. An exact energy correction would additionally require qualified work/density accounting and overlap with the target distribution. Those requirements are independent of a graph-validity improvement. Appendix F retains the global nonequilibrium connection as a conditional extension; the local work teacher is evaluated separately in Appendix G.

REPRODUCIBILITY STATEMENT

Frozen protocols record data-order and generation seeds, source checkpoint hashes, all attempted structures, electronic conditions and oracle costs. Current audits replay 81,600 generated-output source records (including 3,584 FIT records), 1,536 additional projected-coordinate readouts and 31,872 stored source/teacher physical rows; they do not independently rerun optimizers, final neural generation or the oracle. The source-SC and frozen-composition confirmation audits join the moment-control, context, additional-composition, monomer, source-utility, dynamic-attention, product-coordinate and feedback audits are complete. Original-reference calibration is separate from generation and fitting. Implementation tests cover normalization, Jacobians, weighted-tree sampling, symmetry, density gradients and source-law guards. The anonymous release package and final author verification remain outstanding.

AI USE STATEMENT

AI tools assisted research exploration, literature checks, mathematical derivation, implementation, testing, experiment orchestration, analysis and drafting. This is an internal development version; final human-author review remains outstanding.

REFERENCES

- Niclas Dern, Lennart Redl, Sebastian Pfister, Marcel Kollovich, David Lüdke, and Stephan Günnemann. Energy-weighted flow matching: Unlocking continuous normalizing flows for efficient and scalable boltzmann sampling. *arXiv preprint arXiv:2509.03726*, 2025. URL <https://arxiv.org/abs/2509.03726>.
- Leo L. Duan and David B. Dunson. Bayesian spanning tree: Estimating the backbone of the dependence graph. *Journal of Machine Learning Research*, 24(397):1–44, 2023. URL <https://www.jmlr.org/papers/v24/22-0252.html>.
- Ian Dunn and David R. Koes. FlowMol3: Flow matching for 3D de novo small-molecule generation. *arXiv preprint arXiv:2508.12629*, 2025. URL <https://arxiv.org/abs/2508.12629>.
- Jasper Marijn Everink. Random spanning tree markov random field priors for bayesian inverse problems in imaging. *arXiv preprint arXiv:2605.18619*, 2026. URL <https://arxiv.org/abs/2605.18619>.
- Majdi Hassan, Nikhil Shenoy, Jungyoon Lee, Hannes Stark, Stephan Thaler, and Dominique Beaini. ET-Flow: Equivariant flow-matching for molecular conformer generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2410.22388.
- Gerhard Hummer and Attila Szabo. Free energy reconstruction from nonequilibrium single-molecule pulling experiments. *Proceedings of the National Academy of Sciences*, 98(7):3658–3661, 2001. doi: 10.1073/pnas.071034098.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2212.04089>.
- Christopher Jarzynski. Equilibrium free energy differences from nonequilibrium measurements: a master equation approach. *Physical Review E*, 56:5018–5035, 1997. URL <https://arxiv.org/abs/cond-mat/9707325>.
- Junwan Kim, Jiho Park, Seonghu Jeon, and Seungryong Kim. Better source, better flow: Learning condition-dependent source distribution for flow matching. *arXiv preprint arXiv:2602.05951*, 2026. URL <https://arxiv.org/abs/2602.05951>.
- Thomas Kipf, Ethan Fetaya, Kuan-Chieh Wang, Max Welling, and Richard Zemel. Neural relational inference for interacting systems. In *International Conference on Machine Learning*, 2018. URL <https://arxiv.org/abs/1802.04687>.
- Daniel S. Levine, Muhammed Shuaibi, Evan Walter Clark Spotte-Smith, Michael G. Taylor, Muhammad R. Hasyim, Kyle Michel, Ilyes Batatia, Gábor Csányi, Misko Dzamba, Peter Eastman, Nathan C. Frey, Xiang Fu, Vahe Gharakhanyan, Aditi S. Krishnapriyan, Joshua A. Rackers, Sanjeev Raja, Ammar Rizvi, Andrew S. Rosen, Zachary Ulissi, Santiago Vargas, C. Lawrence Zitnick, Samuel M. Blau, and Brandon M. Wood. The Open Molecules 2025 (OMol25) dataset, evaluations, and models. *arXiv preprint arXiv:2505.08762*, 2025. URL <https://arxiv.org/abs/2505.08762>.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.
- Marina Meila and Tommi Jaakkola. Tractable bayesian learning of tree belief networks. Technical Report CMU-RI-TR-00-15, Carnegie Mellon University, 2000. URL <https://publications.ri.cmu.edu/tractable-bayesian-learning-of-tree-belief-networks>.
- Laurence Illing Midgley, Vincent Stimper, Gregor N. C. Simm, Bernhard Schölkopf, and José Miguel Hernández-Lobato. Flow annealed importance sampling bootstrap. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2208.01893.

Suriyanarayanan Vaikuntanathan and Christopher Jarzynski. Escorted free energy simulations: Improving convergence by reducing dissipation. *Physical Review Letters*, 100:190601, 2008. URL <https://arxiv.org/abs/0804.3055>.

Zifan Wang, Alice Harting, Matthieu Barreau, Michael M. Zavlanos, and Karl H. Johansson. Source-guided flow matching. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=p56ZAQUCUr>.

Zhijian Zhou, Junyi An, Zongkai Liu, Yunfei Shi, Xuan Zhang, Fenglei Cao, Chao Qu, and Yuan Qi. Guiding diffusion models with reinforcement learning for stable molecule generation. *arXiv preprint arXiv:2508.16521*, 2025. URL <https://arxiv.org/abs/2508.16521>.

A NORMALIZATION AND AFFINITY-GRADIENT PROOF

For a tree T , let $B_T \in \mathbb{R}^{(N-1) \times N}$ be an oriented incidence matrix and $U \in \mathbb{R}^{N \times (N-1)}$ an orthonormal basis of the centered subspace. Because $B_T \mathbf{1} = 0$, $(B_T U)(B_T U)^\top = B_T B_T^\top$. Cauchy–Binet expresses $\det(B_T B_T^\top)$ as the sum of squares of its N maximal minors. Each minor of a tree incidence matrix is ± 1 , so the determinant is N . The absolute Jacobian in one spatial dimension is therefore $\sqrt{N}$, and in three dimensions it is $N^{3/2}$. Consequently the component density is $q(X \mid T, c) = N^{3/2} \prod_{e \in T} h_e(B_T X)$ and integrates to one. Averaging with Equation (1) yields Equation (3) and preserves normalization. Differentiating the log of each finite positive tree partition with respect to α_e gives its edge-inclusion probability, proving Equation (4). These are applications of standard incidence and tree-ensemble identities.

B SOURCE INVARIANCE UNDER SYMMETRY COUPLING

Let G be the compact group of proper rotations and permutations of identical conditioning labels. Suppose q_0 is G -invariant, $X \sim q_0$, and a pairing algorithm selects $g(X, Y) \in G$ using a target Y . If H is an independent Haar draw, then conditional on the orbit of X and on the selected g , HgX has the uniform orbit distribution of HX , by Haar invariance. Integrating over the unchanged source-orbit distribution recovers q_0 . This argument permits a non-Gaussian source and explains the shared augmentation used in the pairing. It does not imply independence of source and target after pairing or justify the independent-Gaussian velocity-to-score formula.

C COVARIANCE CONTROLS

For $R = \ell \exp(sZ - s^2/2)$ with $Z \sim N(0, 1)$, $\mathbb{E}R^2 = \ell^2 \exp(s^2)$. A uniform direction has second moment $I/3$, so each edge has covariance $vI = \ell^2 \exp(s^2)I/3$. Let A_T map edge coordinates to centered node coordinates. Independence gives $C_T = A_T \text{diag}(v_e)A_T^\top$. Replacing the edge vectors by independent Gaussians of covariance $v_e I$ leaves C_T unchanged and, since all component means are zero, leaves $\mathbb{E}_T C_T$ unchanged. The single-Gaussian control uses a finite, composition-seeded average of these matrices. Averaging this matrix over within-species permutations preserves positive definiteness on the centered subspace and yields species invariance. The Gaussian is exactly normalized for its chosen covariance; equality to the infinite tree-mixture covariance remains approximate.

Additional source-adaptation prior art. Source-Guided Flow Matching already obtains guidance by changing the source under a frozen vector field, including source-space importance and MCMC sampling (Wang et al., 2026). Conditional source learning within flow matching also appears in Kim et al. (2026). Random-tree continuous difference priors have imaging applications (Everink, 2026), and molecular diffusion fine-tuning with physical rewards appears in Zhou et al. (2025). A positive pilot would therefore establish only an internal molecular mechanism result. Comparative utility and cost would still be needed to substantiate the specific structured source adaptation contribution.

D PRESPECIFIED SOURCE-UTILITY PILOT

For each fixed composition and electronic condition c , retain the actual frozen decoder kernel $K_\theta(y | x_0, c)$, including its midpoint integrator and final noise. Changing only latent-tree affinities gives the standard importance identity

$$J(\eta, c) = \mathbb{E}_{q_0 K_\theta} \left[\frac{q_\eta(X_0 | c)}{q_0(X_0 | c)} R(Y, c) \right]. \quad (8)$$

The ratio is a source-space marginal ratio, not the marginal output-density ratio. For M iid cached draws, the leave-one-out centered estimate $\frac{M}{M-1} \frac{1}{M} \sum_i w_i (R_i - \bar{R})$ is unbiased for the improvement over q_0 at a fixed η . Optimizing it on that same bank can overfit.

With common normalized edge kernels, the exact latent-tree divergence is

$$D_T = \sum_e m_{\eta,e} (\log a_{\eta,e} - \log a_{0,e}) - \log \tau(a_\eta) + \log \tau(a_0), \quad (9)$$

where $m_{\eta,e}$ is the prior tree edge marginal. Deploy the mixture $p_{\text{mix}}(T) = (1 - \epsilon)p_0(T) + \epsilon p_\eta(T)$ with $\epsilon = \min(1, \delta/D_T)$, taking $\epsilon = 1$ at zero divergence. Standard KL convexity and data processing through $T \rightarrow X_0 \rightarrow Y$ imply

$$D_{\text{KL}}(p_{\text{mix}}(Y) || p_0(Y)) \leq \epsilon D_T \leq \delta. \quad (10)$$

This controls the raw attempt distribution for a common decoder. It does not certify output density, effective sample size, diversity, validity or equilibrium, and does not apply to earlier experiments that changed decoder weights.

The prospective pilot fixes $\delta = 0.25$ nats, unchanged lognormal edge width 0.2, and utility 0.8 times graph support plus 0.2 times geometric support, retaining all failures. Twenty-four new qualified training-corpus compositions supply 64 bank draws each. Twelve additional compositions are held out of source-head fitting, with four per size bin 8–16, 17–28, 29–40. All 36 originate from the flow’s training corpus; this is not a global pretraining-disjoint test. The actual-utility, within-composition shuffled-utility and coordinate-NLL heads have equal capacity and 30 training epochs, Adam learning rate 0.003, gradient clipping 5 and tree-KL penalty 0.01. Together with the fixed source, all four receive 64 fresh draws per held composition under each of two frozen decoders. The 9216 completed attempts count both bank preparation and validation, and all source draws and structural assays replay. The decoder state tensors remain exactly unchanged.

Completed results. Graph passes out of 768 attempts per arm are 461/462/423/476 for fixed/actual/shuffled/NLL under the first frozen decoder and 410/392/410/409 under the second. Actual-minus-fixed pooled graph gain is -1.11 percentage points, paired 95% interval $[-3.84, 1.76]$ and descriptive composition interval $[-3.58, 0.91]$. Actual-minus-shuffled is $+1.37$ points, with intervals $[-1.63, 4.36]$ and $[-1.11, 3.91]$. The development gate fails. Distinct connectivity sums are 421/423/377/429 and 373/352/377/375. Neither a repeated learned increment nor broad failure of all utility-trained sources follows from this small pilot. The actual-head FIT-bank utility estimates improve by 0.0262/0.0220, while shuffled heads improve their own shuffled objectives by 0.0205/0.0154. These fitted-bank readouts do not predict a repeated fresh benefit and do not uniquely distinguish finite-bank overfitting, composition shift and source-family limitations.

Preparing each actual-utility model requires at least 462.4/473.7 seconds for bank generation plus source training. The full four-arm jobs complete in 24:02/24:43. The separate corpus-qualification stage was not timed, so the preparation numbers are lower bounds. No new quantum-energy queries were made. Both the earlier negative studies and this failed gate remain part of the evidence.

E FURTHER CONTROLLED ARCHITECTURE TESTS

Current-geometry structure. A differentiable tree-marginal residual adapter was compared with independent edge mass and fixed tree affinities, under two newly matched 3000-update training seeds and 12 reused monomer development conditions. Graph passes out of 768 are 316/321/322/318 and 312/300/298/295 for no block, learned tree, fixed tree affinities and learned local mass. Learned-tree

minus no block is -0.46 percentage points, paired 95% interval $[-1.95, 1.11]$. Learned-tree minus local is $+0.52$ points, paired interval $[-0.59, 1.56]$ and descriptive composition interval $[0.07, 1.04]$. The gate fails; the favorable composition interval is not selected in isolation. A relative-edge-floor gradient defect was corrected before this final comparison. The 1472 saved outputs of the interrupted implementation are preserved separately; only its unaffected complete no-block baselines were reused.

Contact connectivity is insufficient. A further prototype sampled a normalized coordination-sensitive heavy-core tree and terminal allocations, and generated bounded tree-edge radii with spherical directions. A product-space flow preserves contact connectivity by construction. The matched Cartesian flow, its deterministic final radial projection, and the trained product flow achieve graph counts 311/202/165 and 295/193/156 out of 768. Both constrained outputs have zero disconnected cases, but product-flow graph validity falls by 18.55 points versus Cartesian flow, paired 95% interval $[-20.83, -16.28]$, and by 4.82 points versus final projection, $[-6.90, -2.73]$. Thus correct geometric support does not establish chemical validity or useful learning. The 3072 neural outputs and 1536 derived projections are all retained.

Self-conditioning and unlabelled relation feedback. Earlier source experiments used the explicitly declared memoryless one-pass coordinate path, not the full native FlowMol3 sampling algorithm. For a linear FM displacement head, $\hat{X}_1 = X_t + (1 - t)v_0$ supplies the appropriate endpoint estimate to a geometry feedback branch. Two differentiable passes, supervised by their mean velocity-FM error, preserve this parameterization. Additional variants feed soft edge codes from the first pass or their within-molecule average to the second pass. Only coordinate error trains those codes; no bond-order label or loss is added, and no chemical semantics are assigned to the four channels. Learning latent interactions is established prior art (Kipf et al., 2018).

The one-pass, geometry-only, pair-code and pooled-code models produce graph counts 363/417/422/432 and 312/342/334/333 out of 768. All use 128 primitive denoiser calls per sample; feedback uses 32 midpoint steps with two passes, versus 64 single-pass steps. Training updates match, but feedback training compute is higher. Geometry feedback improves graph support by 5.47 points, paired 95% interval $[3.06, 7.94]$ and descriptive composition interval $[4.10, 6.77]$. This is a known-technique baseline improvement. Pair codes add -0.20 points versus geometry, $[-1.50, 1.17]$, and -0.59 versus pooled codes, $[-2.02, 0.85]$. The relation-learning gate fails, despite verified coordinate-only gradients and changes in the edge head. No new AI contribution is established by the self-conditioning gain alone.

All these controls retain the bond-free interface, original electronic metadata, zero bond loss and the declared train/evaluation separation. The later 12-condition panel is reused development data; it is not an untouched confirmatory test. No new quantum-energy queries were made in these three architecture studies.

Internal geometry versus accumulated velocity. A final controlled hypothesis separated the velocity accumulator V from the internal geometry used by coordinate-updating layers. The tested geometry was either fixed at X_t or set to $X_t + (1 - t)V$, while the output remained the displacement head $X_t + V$. The standard endpoint/velocity identity motivates this inductive bias but does not prove that arbitrary learned internal geometries are invalid. No network parameters were added. Graph counts for original/geometry-SC/fixed/time-scaled geometry were 384/410/167/328 and 327/369/156/315 out of 768. Time-scaled minus original geometry is -4.43 percentage points, paired 95% interval $[-6.84, -2.02]$. Time-scaled minus fixed geometry is $+20.83$ percentage points, paired 95% interval $[+18.29, +23.44]$. Time-scaled minus geometry self-conditioning is -8.85 percentage points, paired 95% interval $[-11.46, -6.25]$. The internal-geometry gate fails. All 6144 source/output records and matching parameter counts, data and checkpoint recipes were audited. The original and geometry-SC references reuse their earlier trained checkpoints with new source streams; their 4.43-point SC difference is not an extra model-training replicate. These results retain the known self-conditioning improvement and do not establish a new useful architecture or a calibrated energy distribution.

F CONNECTION TO JARZYNSKI AND ESCORTED TRANSPORT

The motivating connection to nonequilibrium force measurements is retained, but is separate from the evaluated validity experiment. Jarzynski relates exponential work averages to equilibrium free-energy differences under a qualified nonequilibrium protocol; Hummer–Szabo reconstruction applies this principle to single-molecule pulling (Jarzynski, 1997; Hummer & Szabo, 2001). Artificial escort flows generalize the transport protocol while retaining the appropriate work accounting (Vaikuntanathan & Jarzynski, 2008). None of these identities is a new result of this paper.

A change-of-variables version clarifies what our normalized source could supply. Fix a composition and intrinsic coordinate measure. Let $\rho_1(x) = \mathbf{1}_\Omega(x) \exp[-\beta E(x)]$ be a target with finite, nonzero $Z_1 = \int_{\mathcal{H}_N} \rho_1(x) dx$, for example a declared confined molecular ensemble. Normalizability must not be assumed for dissociating fragments in unbounded space. Let $q_0 > 0$ be the normalized source in Equation (3), and let $\Phi : \mathcal{H}_N \rightarrow \mathcal{H}_N$ be a differentiable bijection with nonzero intrinsic Jacobian. For $\Phi(x_0) \in \Omega$, define the generalized work

$$W_\Phi(x_0) = E(\Phi(x_0)) + \beta^{-1} \log q_0(x_0) - \beta^{-1} \log |\det D\Phi(x_0)|, \quad (11)$$

and set $W_\Phi = +\infty$ otherwise. Then

$$\mathbb{E}_{q_0}[e^{-\beta W_\Phi}] = \int_{\mathcal{H}_N} \rho_1(\Phi(x_0)) |\det D\Phi(x_0)| dx_0 \quad (12)$$

$$= Z_1. \quad (13)$$

Equivalently, the auxiliary source energy $E_0 = -\beta^{-1} \log q_0$ has $Z_0 = 1$ in the declared measure, yielding a Jarzynski-type normalizer identity. The Jacobian term accounts for compression by the escort map; the work cannot be replaced by an endpoint energy difference alone. Weighted observables require the same complete weight. Even a correct identity need not give an efficient finite-sample estimator when the weights have poor overlap.

For a sufficiently regular exact ODE flow, the intrinsic log determinant is the integral of the intrinsic velocity divergence. Our measured midpoint sampler followed by terminal Gaussian noise is not that exact diffeomorphism. A valid implementation would need either its own qualified map/Jacobian or a specified forward/reverse stochastic path law that includes the noise and every correction. We have not performed this global reweighting for the source-SC results. A separate local anchored work teacher is defined and tested in Appendix G. No global work-variance, free-energy, effective-sample-size or Boltzmann-distribution claim is inferred from their graph or geometry pass rates.

G LOCAL FORCE-ESCORTED WORK DISTILLATION

This separate experiment brings energy/force feedback into training while keeping the positive source-comparison checkpoints frozen. Energy-weighted flow matching and weighted physical-sample training already exist (Dern et al., 2025; Midgley et al., 2023); escorted work identities are also established (Vaikuntanathan & Jarzynski, 2008). Our local construction is an application of these principles, not a new global Boltzmann-sampling theorem.

A specified local reference and target. For a generated FIT anchor A with accepted perceived graph G_A , define $q_A(X) = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. At $kT = 0.0258519998$ eV (300 K), remove net force and set

$$\sigma_A = \min \left(0.03 \text{ \AA}, \sqrt{\frac{0.10 \text{ \AA} kT}{\|F_+(A)\|}} \right), \quad \delta_A = \frac{\sigma_A^2}{kT} F_+(A). \quad (14)$$

A vanishing force selects the maximal width. These quantities are fixed before drawing $X \sim q_A$. The escort $Y = X + \delta_A$ is a translation on the centered subspace and has determinant one. The local unnormalized target is

$$\rho_A(Y) = q_A(Y) \exp \left[-\frac{E_+(Y) - E_+(A)}{kT} \right] \mathbf{1}_{G_A}(Y), \quad (15)$$

where the support requires the same labelled perceived bond orders, formal charges and radical counts, as well as the existing connected/nonoverlap assay. These are derived structural checks, not supplied bond labels or a bond-classification loss. This target includes a Gaussian restraint around A and does not equal the unrestricted molecular Boltzmann law.

Table 4: Direct physics-distillation graph support per 640 attempts. The frozen reference is not refitted; other arms have identical additional update budgets.

	Continuation	Frozen	Replay	Unweighted escort	Complete work
1		418	340	330	344
2		377	345	371	362

Complete local work. With reference Hamiltonian $H_0 = -kT \log q_A$ and target Hamiltonian $H_1 = H_0 + E_+ - E_+(A)$ plus hard support, the generalized quench work is

$$W_A(X) = E_+(Y) - E_+(A) + kT[\log q_A(X) - \log q_A(Y)] \quad (16)$$

on valid support, and $+\infty$ otherwise. Change of variables gives $\mathbb{E}_{q_A}[\exp(-W_A/kT)] = \int \rho_A(Y) dY$. The two reference-density terms correct for the escort’s proposal shift. Energy difference alone would omit them. Two analytic tests verify constant work for a constant-force escort and the normalizer/weighted mean for a quadratic energy. These tests validate the local arithmetic, not a global molecular sampler.

Training and controls. Each continuation selects eight distinct 17–28-atom TRAIN compositions by a fixed metadata hash and generates 32 new FIT anchors per composition. They are excluded from the evaluated compositions. Invalid anchors are recorded rather than replenished. Each valid anchor produces eight escorted particles. Particles outside its declared support receive zero weight. The work arm samples training endpoints using self-normalized $\exp(-W_A/kT)$ weights; the unweighted escort arm samples the same valid particles uniformly. A replay arm uses the retained raw anchors, and a frozen model supplies the reference. All-zero-particle anchors are recorded and excluded identically from all student pools.

Each student alternates 500 original-reference FM updates with 500 generated-FIT endpoint updates, sharing initialization, anchor draws, optimization and sampling budgets. Physical preparation costs differ: replay needs no force queries, the unweighted escort needs anchor forces, and complete work additionally needs energies at valid proposed endpoints. Actual shared research cost and these method requirements are reported separately. No oracle is queried by the final generator. Physics enters through training targets. The generator retains its legacy constant temperature input; this is not a model trained to predict different temperatures. Finite-particle normalization biases the teacher relative to the exact normalized local target. Its ESS refers only to eight local particles, not global generator outputs. Original-reference mixing and unknown chemical-isomer weights further prevent a global equilibrium claim, even if the local identity is exact.

Direct-distillation outcome. Both continuations finish all three 1,000-update student fits. The 512 generated FIT anchors yield 351 graph-supported anchors, of which 349 have at least one supported escorted particle. All original draws and unsupported particles remain in the records. Mean local particle ESS by composition is about 1.27–1.87 out of 8, indicating substantial weight concentration and limited finite-particle resolution of the local target.

On paired common-graph support, complete work versus the unweighted escort reduces mean energy by 0.00758 eV/atom (conditional paired 95% interval $[-0.01424, -0.00113]$ for the signed difference) and force RMS by 0.16174 eV/Å (interval $[-0.27660, -0.05590]$). Both continuation signs favor complete work. All-output readouts agree in direction. Thus an incremental physical-quality signal is present relative to the matched escort student.

However, all direct continuations harm the frozen model. Complete work has graph support 55.16%, versus 62.11% for frozen, and common-supported energy is *higher* by 0.10587 eV/atom (interval $[0.09009, 0.12168]$). The prespecified adoption gate therefore fails. The matched physics increment does not establish an overall generator improvement. The replay control shows that plain continuation on these mixed targets already produces substantial regression; it does not uniquely identify the cause of that regression.

All 512 FIT source draws, 5,120 evaluation outputs, local work values, support masks, selections, checkpoints and physical readouts are audited. Teacher preparation uses 6,232 raw energy/force queries: 702 at anchors and 5,530 at supported particles. An unweighted escort could omit the

Table 5: The single parameter-arithmetic test preserves most observed graph support while improving the eSEN readout. Energy differences are relative to frozen on paired common-graph support, with conditional paired 95% intervals. This test was designed after the direct-continuation result on the same panel.

Model	Graph passes (two streams)	Energy change (eV/atom)
Frozen	418/640; 377/640	0
Escort difference	407/640; 387/640	-0.01554 [-0.02084, -0.01031]
Work difference	406/640; 382/640	-0.01604 [-0.02188, -0.01013]

particle-energy queries; replay could omit all physical queries. Evaluating the four generated-output arms uses another 10,240 queries, reported separately from inference cost. No global Boltzmann, output-ESS, work-law novelty or successful direct-distillation claim follows from this result.

Separating the paired update from continuation regression. The failed direct continuation motivates one subsequent fixed-coefficient parameter-arithmetic test. Parameter arithmetic is established (Ilharco et al., 2023); we make no claim to invent it or to subtract functional drift exactly. For the same initialization and paired training draws, construct

$$\theta_{\text{new}} = \theta_{\text{frozen}} + (\theta_{\text{physical}} - \theta_{\text{replay}}), \quad (17)$$

for both unweighted-escort and complete-work students. The coefficient is one; there is no coefficient, layer-selection or extra-training sweep. Only parameter tensors change, while all buffers and the source distribution remain fixed. All four reconstructed checkpoints are checked tensor by tensor and restore.

Both continuation signs improve for energy and force in both transferred variants. The work-difference variant reduces force RMS by 0.38403 eV/Å on common graph support (interval [-0.48124, -0.28932]); all-attempt energy and force readouts also improve. Its observed graph-rate difference is -0.55 percentage points, with interval [-2.58, 1.56]. The frozen gate used a two-point tolerance on the *observed pooled rate*; this is not a formal statistical noninferiority certificate, and a larger loss remains compatible with the interval.

The additional effect of complete work over the transferred unweighted escort is not established: common-supported energy difference is -0.00243 eV/atom, with interval [-0.00756, 0.00264], while all-attempt energy difference is +0.00022 with interval [-0.00557, 0.00649]. Force and graph intervals also span zero. Consequently the practical physical-quality signal does not require an asserted Jarzynski-specific advantage. The unweighted variant would require only 702 anchor queries for teacher preparation, versus 6,232 for the complete-work variant. Both use the same single-network-128-call inference budget.

This exploratory intervention passes the frozen-baseline physical-improvement gate for work transfer, but fails the separate extra-work-over-escort gate. It adds 2,560 generated outputs and 5,120 physical queries. No new optimizer steps are used in the transfer itself. Existing panel outcomes informed the choice of intervention, so independent composition confirmation and physical-model validation remain necessary. These results support investigating the simpler physics update while retaining complete work as a defined and tested comparison; they do not establish global Boltzmann sampling or new task-arithmetic theory.